



COSMO INSTRUMENTS (I) PVT. LTD.
SPECIALIST IN AIR LEAK AND FLOW TEST



AIR LEAK TESTER
MODEL LS-R900

ERROR AND TROUBLESHOOTING MANUAL

1 Error

1.1 When an Error Occurred

The error code is displayed when an error occurs.

The error description, probable causes and the treatments are displayed by tapping the displayed code.

1.2 Error List

Menu to view descriptions, probable causes and treatments and all the errors.

Go to: Troubleshooting > Error List

The errors are divided every 10 codes.

Tapping $\updownarrow$ goes forward or back within 10 errors.

2 Error Messages and Treatments

2.1 ERROR 1: PS Offset Error

Timing: During power-on check procedure

Criteria: Pressure sensor (PS) offset exceeds $\pm 2\%$ of its range.

Probable Cause	Treatment
Test pressure sensor (PS) offset is out of tolerance when the power is turned on.	Adjust the PS offset. Go to: Maint. > Inspection > Sensor > PS(P1) Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.

2.2 ERROR 2: PS Out of Range

Timing: At the end of Pressurization (CHG) or Precharge (PCHG) stage

Criteria: Test pressure exceeds the sensor range in CHG or PCHG stage.

Probable Cause	Treatment
Test pressure sensor (PS) was pressurized exceeding the sensor full-scale.	Adjust the test pressure. Pay extra attention for low pressure models.
Test pressure sensor (PS) offset is out of tolerance.	Adjust the PS offset. Go to: Maint. > Inspection > Sensor > PS (P1)

Cable disconnection or malfunction of the test pressure sensor (PS)	Adjust the PS offset. Go to: Maint. > Inspection > Sensor > PS(P1) Contact your local Cosmo representative if the offset exceeds $\pm 2\%$ of the sensor range.
Malfunction of the test pressure sensor (PS)	Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.

NOTE When the error occurred in PCHG, stages only in grey areas are applicable. When it occurred in CHG stage, the stages in shadowed areas are also applicable as well.

2.3 ERROR 3: Test Pressure Error

Timing: Test pressure too low: At the end of Pressurization (CHG)

Test pressure too high: Always monitored

Precharge pressure too low: At the end of Precharge (PCHG)

Precharge pressure too high: Always monitored

Criteria: Test pressure exceeds upper or lower limit in CHG or PCHG stage.

Probable Cause	Treatment
"0" is set to the lower limit	Set a numerical number other than "0" to the lower limit.
Upper and Lower limits for Test pressure or Precharge are too close or inappropriate.	Set larger limits. For test pressure limits: Go to: Settings > Advanced Settings > Test Press > Upper Press Limit / Lower Press Limit For Precharge limits: Go to: Settings > Advanced Settings > CHG Support > Precharge Upper Limit / Precharge Lower Limit
Pressurization time is insufficient. (When pressure is lowered.)	Extend CHG timer. Go to: Settings > Advanced Settings > Timer > Charge (CHG)
Precharge time is insufficient. (When Precharge pressure is lowered.)	Extend PCHG timer. Go to: Settings > Advanced Settings > CHG Options > Precharge timer
Fluctuation or a drop in the source pressure	Check the source pressure or the regulator setting. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Leaks from seals, part and fittings	Check the seals, part and fittings for possible leaks.

Malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.
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NOTE When the error occurred in PCHG, stages only in grey areas are applicable. When it occurred in CHG stage, the stages in shadowed areas are also applicable as well.

2.4 ERROR 4 BAL1 Lost Test Pressure

Timing: At the end of Equalization (BAL1)

Criteria: Programmed Test pressure lower limit

Probable Cause	Treatment
Leaks from seals, part and fittings	Check the seals, part and fittings for possible leaks.
Fluctuation or wrong setting of the pilot pressure	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Malfunction of the solenoid valve, SV4, or air-operated valve, AV3.	Contact Cosmo for repair.

2.5 ERROR 10: DPS Offset Error

Timing: During power-on check procedure

Criteria: Differential Pressure sensor (DPS) offset exceeds $\pm 3\%$ of its range.

Probable Cause	Treatment
Differential pressure sensor (DPS) offset is out of range when the power is turned on.	Adjust the DPS offset. Go to: Maint. > Inspection > Sensor > DPS Contact Cosmo for repair if the offset exceeds $\pm 30\%$.

2.6 ERROR 11: Air Operated Valve Error 1

Timing: At the end of PCHK

Criteria: Test pressure sensor (PS) offset exceeds $\pm 1\%$ of the sensor range.

Probable Cause	Treatment
Pilot pressure is not stable or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Test pressure sensor (PS) offset exceeds $\pm 1\%$ of the sensor range.	Adjust the PS offset or enable Auto-Zero feature to reset the pressure residue of the previous test. PS Offset: Go to: Maint. > Inspection > Sensor > PS(P1) PS Auto-Zero: Go to: Settings > Advanced Settings > Test Press > PS Auto-Zero
Charge Delay (DL1) timer is too short	Set the DL1 timer to 0.2 s or longer. Go to: Settings > Advanced Settings > Timer > Charge Delay (DL1)
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.7 ERROR 12 Air Operated Valve Error 2

Timing: At the end of Pressurization (CHG) or Precharge (PCHG) stage

Criteria: Auto-zero of PS is smaller than 1% of the sensor range at the end of CHG

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Pressure source is disconnected.	Check the pressure source and the regulator setting.
Test pressure is too low for high pressure models, H20 and H49.	Adjust the test pressure within the test pressure range.
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

NOTE When the error occurred in PCHG, stages only in grey areas are applicable. When it occurred in CHG stage, the stages in shadowed areas are also applicable as well.

2.8 ERROR 14 Air Operated Valve Error 4

Timing: At the end of Air-Blow (BLW)

Criteria: The difference pressure during Air-Blow did not reach the Blow ΔP Limit

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Air-Blow (BLW) timer is too short or Blow ΔP Limit is too high.	Extend the Air-Blow (BLW) timer or lower the Blow ΔP Limit. Air-Blow (BLW) Timer: Go to: Settings > Advanced Settings > Timer > Air-Blow (BLW) Blow ΔP Limit: Go to: Settings > Advanced Settings > Self Check > Blow ΔP Limit
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.9 ERROR 15: Air Operated Valve Error 5

Timing: At the end of Stabilization (BAL2). Only for High press and External Press.

Criteria: Pressure switch monitoring the pilot pressure for the Balance (BAL) valve is not activated.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source of the LS-R900. Setting up a dedicated pressure source for the LS-R900 is recommended.
Malfunction of the pressure switch monitoring the pilot pressure for the Balance (BAL) valve.	Contact Cosmo for repair. As a provisional measure, the pressure switch monitoring can be disabled. Go to: Settings > Common Settings > Special > PSW Monitoring > Disable

2.10 ERROR 16: Air Operated Valve Error 6

Timing: During idle state

Criteria: DPS offset exceeded the Idle ΔP Check limit within the programmed Idle ΔP Check Time

Probable Cause	Treatment
DPS offset exceeded its monitoring limit while the LS-R900 is in idle state.	Adjust DPS offset. Go to: Maint. > Inspection > Sensor > DPS Contact Cosmo for repair if the DPS offset exceeds $\pm 30\%$ of its range.
Exhaust time is insufficient.	Extend Idle ΔP Check Timer or Exhaust timer. Idle ΔP Check Timer: Go to: Settings > Advanced Settings > Self Check > Idle ΔP Check Timer Exhaust timer: Go to: Settings > Advanced Settings > Timer > Exhaust (EXH)
Malfunction of the fill valve: SV1 or AV1	Contact Cosmo for repair.

2.11 ERROR 17: Blockage Check Error

Timing: At the end of Pressurization (CHG)

Criteria: The Blockage data exceeded the programmed tolerance sampled Blockage Check data in the Memory.

Probable Cause	Treatment
Something is blocking the air way of the external pneumatic circuits (valves).	Check the external pneumatic circuit (valves).

2.12 ERROR 21: DPS Stopped Oscillating

Timing: Always monitored

Criteria: DPS stopped oscillating.

Probable Cause	Treatment
Malfunction of the DPS or power source or cable disconnection	Contact Cosmo for repair.

2.13 ERROR 22: Stop Valves Closed

Timing: At the end of PCHK

Criteria: The stop valve monitoring switch is ON/OFF

Probable Cause	Treatment
Stop valves of WORK and MASTER ports are closed, which disturb the cover from closing. (The stop valve monitoring switch is not pressed.)	Open the stop valves.
If the error occurs even though the stop valves are opened, the stop valve monitoring switch may be malfunctioned.	Contact Cosmo for repair.

2.14 ERROR 23: Mastering Error

Timing: At the end of the last iteration of DET for Mastering value sampling

Criteria: Leak data at the end of the last DET iteration exceeded the Mastering Limit in the Mastering Sampling.

Probable Cause	Treatment
Pressurization and stabilization time is insufficient.	Extend Charge (CHG) and Balance (BAL2) timers. Go to: Settings > Advanced Settings > Timer > Charge(CHG) / Balance(BAL2)
MB1 (Mastering Equalization) timer, MB2 (Mastering Stabilization) timer and/or Mastering Iterations are inappropriate.	Check each setting. Go to: Comp. > Mastering Settings > Basic Ref: The recommended settings for MB1 and MB2 timers are 0.5 s. Make sure that the last DET data is not a negative figure.
Upper and Lower limits for Mastering are inappropriate.	Set larger limits. Go to: Comp. > Mastering Settings > Basic > High Limit / Lo Limit Ref: Typically the Mastering limits are set to be about 120 to 150% of the 1st DET in a Mastering value sampling. Default: ± 250 [Pa]

2.15 ERROR 24 K(Ve) Value Out of Range

Timing: The last DET in K(Ve) Automatic Setup

Criteria: Calculated K(Ve) exceeded 100L.

Probable Cause	Treatment
The current K(Ve) settings does not match the calibrator used for K(Ve) Automatic Setup causing the measured value exceeding 100L.	Check the settings for the calibrator. Go to: K(Ve) > K(Ve) Settings > Basic ALC: ALC Displacement or ALC Reading Leak Master: LM Flow [mL/min]

2.16 ERROR 25 Leak Limit Out of Range

Timing: The last DET in K(Ve) Automatic Setup

Criteria: K(Ve) Leak limits exceeded the DPS range after K(Ve) Automatic Setup

Probable Cause	Treatment
Leak limits exceeded the DPS range after executing the K(Ve) Automatic Setup.	Change the Leak unit to a pressure unit and perform K(Ve) Automatic Setup again. Go to: Settings > Advanced Settings > Unit > Leak

2.17 ERROR 51 to ERROR 61: System Errors

Usually system errors (ERROR 51 to ERROR 61) are caused by malfunction of electrical components.

NOTE ERROR 51 (Lo Battery SRAM Error) may occur because of the battery life.
System Errors

Error Code	Error Message	Description
ERROR 51	Lo Battery SRAM Error	Probable Cause 1: The battery has discharged completely. Treatment 1: If this error occurs, none of the measurements is enabled. Replace the battery instantly. Probable Cause 2: If the same error occurs after replacing the battery, some electrical parts may be malfunctioned. Treatment 2: Contact Cosmo for repair after executing a System Back up.
ERROR 52	SPI2-res	AD Communication failure

ERROR 53	SPI1-res	I/O Communication failure
ERROR 59	Flash data area bad track Error	
ERROR 60	Flash program area WR Error Kernel	
ERROR 61	RAM checksum Error	

Contact Cosmo for repair after executing System Backup.

3 Large Leak List

Display	Probable Cause	Treatment
CHG Large Leak WORK side CHG Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
DL2 Large Leak WORK side DL2 Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Precharge (PCHG), Pressurization (CHG) or Equalization (BAL1) timer. Go to: Settings > Advanced Settings > CHG Options > Precharge Timer (PCHG) Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Equalization (BAL1)
BAL2 Large Leak WORK side BAL2 Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) or Equalization (BAL1) timer. Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Equalization (BAL1)
DET Large Leak WORK side DET Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the seals, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) or Stabilization (BAL2) timer. Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Stabilization (BAL2)

If the problem persists without identifiable causes, please conduct the No Leak Check.

1) Close both the WORK and MASTER stop valves on the rear panel of the tester.

2) Go to: Maint. > Inspection > Leak Check > No-Leak Check

Contact Cosmo for repair, if internal leak is found.

4 Frequent (+) Fails

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

4.1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the LS-R900 is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

4.2 Check the fixture condition.

Probable Cause	Treatment
Leaks from tube fittings	Look for leaks in the fittings by performing a bubble test applying soap solution. Redo the tubing if needed.
Deformation of tube	Replace the tube with the harder one that does not deform with the air pressure.

Proceed to the next item to check if the problem persists without identifiable causes.

4.3 Check the sealing condition.

Probable Cause	Treatment
Sealing material is missing.	Place the sealing material.
Sealing surface is contaminated.	Clean the sealing surface.
Sealing material is damaged or worn-out.	Replace it with a new one.
Sealing deforms when the fixture clamps.	Check the following: – Whether the clearance between the sealing material and the groove is enough. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – If the thrust force of the cylinder has lowered.

Proceed to the next item to check if the problem persists without identifiable causes.

4.4 Check whether there were environmental changes.

Probable Cause	Treatment
Tested part is exposed to the direct wind from air conditioner or fan.	Move the source of the wind to where the wind does not directly hit the tested parts.
Some air tools are branched off the pressure source for the LS-R900 causing fluctuation in the pressure source.	Avoid using air tools branching off the pressure source for the tester to supply a stable air. Setting up a dedicated pressure source for the LS-R900 is recommended.
Air compressor capacity is insufficient.	Use the air compressor whose capacity is large enough.
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.

Proceed to the next item to check if the problem persists without identifiable causes.

4.5 Check the condition of the tested parts.

Probable Cause	Treatment
Part temperature is higher or lower than the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line.
The tested parts are wet.	Improve the drying process or add one if there isn't any.
The tested parts get deformed by pressurization.	Add a stopper to prevent the deformation.
Leak due to the gas porosity or internal leak	Look for leaks in the fittings by performing a bubble test applying soap solution. Redo the tubing if needed. If no leak is confirmed, there may be internal leaks. If there is a leak, re-evaluate the production process.

5 Frequent (-) Fails

There are two types of the causes for the negative fails. One is caused by a pressure rise in WORK-side circuit and the other is caused by a pressure reduction in the MASTER side circuit.

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

5.1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the LS-R900 is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

5.2 By a pressure rise in WORK-side circuit

Probable Cause	Treatment
Sealing is not stable.	Check the following: – Whether there is enough clearance between the sealing material and the groove. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – Whether the thrust force of the cylinder is too high.
A rise in temperature of the air inside the tested part due to the temperature rise of the cold tested part trying to match the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line. If the part is wet, add or improve the drying process.

Proceed to the next item to check if the problem persists without identifiable causes.

5.3 By a pressure reduction in MASTER side circuit

Probable Cause	Treatment
There are leaks from the Master or the fittings on the MASTER side.	Check Master part and the fittings for possible leaks. Replace the Master part and fittings if leaks are found.
Deformation of MASTER side tube	Replace it with tube that is rigid enough not to deform.

Adiabatic compression effect of the Master	The size of the Master Chamber may be wrong, or the BAL2 timer may be too short. Replace the master to the one with good temperature stability. Extend BAL2 timer if possible.
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Proceed to the next item to check if the problem persists without identifiable causes.

5.4 By over compensation

Probable Cause	Treatment
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.